

Parameter	Initial value
Minimum contiguous buildable area	20 acres
Maximum screening slope	15%
Stream/wetland screening buffer	100 feet

Find candidate land

Order	Geoprocessing Tool	Input → Output / setting
1	Select Layer By Attribute	Parcels → Parcels_Size; select parcels above an initial gross acreage threshold. Start above 20 acres because exclusions will reduce usable area.
2	Select Layer By Location	Parcels_Size + eligible zoning/land-use areas → Parcels_Zoning; identify existing industrial or other districts you decide to evaluate. Preserve parcels in other jurisdictions as a separately labeled category if doing a countywide policy study.
3	Select Layer By Location	Parcels_Zoning + water service areas → Parcels_Water; identify parcels with access to public water service.
4	Slope	Bare-earth DEM → Slope_Pct; output measurement: Percent rise.
5	Reclassify	Slope_Pct → Slope_Eligible; cells at or below 15% = 1, steeper cells = NoData.
6	Raster To Polygon	Slope_Eligible → Slope_Eligible_Polygon; dissolve polygons if needed.
7	Pairwise Buffer , then Merge/Dissolve	Buffer streams and wetlands; combine these with flood hazard areas, protected lands, county parks, state parks, agricultural security areas, natural areas, and game lands → Exclusion_Areas. Maintain a separate reason field or separate source layers so exclusions remain explainable.
8	Pairwise Intersect	Parcels_Water + Slope_Eligible_Polygon → Parcel_Gentle_Land.
9	Pairwise Erase	Parcel_Gentle_Land minus Exclusion_Areas → Buildable_Pieces.
10	Multipart To Singlepart	Buildable_Pieces → Contiguous_Acreage; separates multipart parcel features that were created by erasing exclusion areas into singleparts.
11	Calculate Geometry Attributes	Contiguous_Acreage → field 'build_ac' (data type: Double, number format: Numeric + 'Rounding' set to 2 decimal places); calculate acres for each contiguous piece.
12	Select Layer By Attribute	Field 'build_ac' >= 20 → Contiguous_Acreage_20AC; prevents several disconnected small pieces from being counted as one usable site.
13	Pairwise Dissolve	Contiguous_Acreage_20AC → Candidate_Footprints; dissolve by 'taxidnumbe' field and Sum the build_ac field to get the total buildable acreage for a multipart parcel.
14	Join	Candidate_Footprints to original parcels → Candidate_Parcels; retain parcel ID and the largest contiguous BUILD_AC.
15	Spatial Join	Candidate_Parcels + CCC_Parcels → Candidate_Parcels_Ver_1; identifies which Climate Change Connectivity priority zones that parcels are in.
15.1	Select Layer By Attribute	Candidate_Parcels_Ver_1 → Candidate_Parcels_Ver_1A; select parcels that are in Very Low or Low 'Regional Connectivity Priority' zones.

15.2	Select Layer By Attribute	Candidate_Parcels_Ver_1 → Candidate_Parcels_Ver_1B; select parcels that are in Very Low, Low, or Medium 'Regional Connectivity Priority' zones.
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